Drone Controls Summary

DJI Tello control inputs in the Drone Race Hacks toolkit

The drone is flown with four control inputs, the same four "sticks" as the Tello SDK **rc** command. The repo has no joystick or keyboard control: the flight engine and the simulation policy generate the inputs automatically.

The four manipulators

The engine sends `rc <right> <forward> <up> <yaw>` over UDP to the Tello at port 8889, about 20 times per second (`core/tello_dual_video.py:115`). Each channel is an integer from -100 to 100.

Channel	Does	Positive means	Roughly at +/-100
right	Strafe sideways (roll)	Right	about 0.8 m/s
forward	Move along the nose (pitch)	Forward	about 0.8 m/s
up	Change altitude (throttle)	Climb	about 0.6 m/s
yaw	Rotate around the vertical axis	Clockwise	about 60 deg/s

Speeds come from the motion model in `predict_tag` in `tools/tello_policy.py`.

Discrete commands

- **takeoff** and **land**: blocking SDK commands, each with a 15 s timeout.
- **streamon**: starts the video stream on connect.
- Takeoff happens about 20 s after start.
- The drone lands automatically when centered at the target distance, or on reconnect after a link drop (`LAND_ON_LOSS`).

How the sticks are set

The PID controller in `tools/tello_policy.py` sets the sticks from the AprilTag pose:

Stick	Behavior	Gains (kp, ki, kd)	Limit
yaw	Turn toward the tag	YAW_PID = (2.0, 0, 0.15)	+/-35
right	Strafe onto the lane line	LATERAL_PID = (40, 0, 6)	+/-25
forward	Close in or back off to the standoff distance (LAND_RANGE_M = 1.3 m)	DIST_PID = (10, 0, 3)	+/-25
up	Only before the tag is first seen: climb of 22 up to a 2 m ceiling. Zero afterwards.	none	22

Policy modes

- **wait / ground**: all sticks at zero.
- **climb**: *up* only, until a tag is seen.
- **aim**: yaw only, so the nose faces the tag before the drone translates.
- **lane / track / back**: strafe and move forward or back.
- **land**: all sticks at zero, then the land command.
- **search**: if the tag is lost for 0.4 s, stop and sweep yaw left and right.

Simulation

In Isaac Lab (`AS-Tello-Race-v0`) the action is the same four rc channels, scaled to [-1, 1] and passed through `TelloVelocityController`. The PPO policy in `simulation/` learns to drive the same four inputs.

Emergency stop

- Flip the drone upside down, which cuts the motors.
- Press its power button.
- Kill the software; with no commands arriving, the Tello auto-lands after about 15 s:

```
pkill -f tello_link.py ; pkill -f tello_dual_video.py
```

Sources: README.md, FLYING.md, tools/tello_policy.py, core/tello_dual_video.py, simulation/README.md. The simulation environment code was only skimmed.